Agentic MT: Phases 1–6 Update

Where the theory stands

Six phases in, the central joint-necessity claim (slow thinking and tool use are each insufficient alone, jointly sufficient, and most valuable for low-resource pairs) has not been confirmed or falsified — every attempt to test it directly has run into a model confound before producing a clean signal. What we do have is real, independent evidence against the simplest instantiations of two supporting mechanisms (QE-in-the-loop, source-side triggering), and a new, well-evidenced finding that a base model's raw pretraining exposure to a language is itself a hard gate on generation, separate from anything the agentic scaffolding does. Net effect: two of the five proposed research directions have real evidence against their first-pass implementations, the flagship joint-necessity test remains open, and we've learned something the theory didn't originally account for.

Findings, by implication

Challenges to specific proposed mechanisms

- **QE-in-the-loop (Sec. 5.3) is not viable as specified for most of the target languages.** COMET-KIWI vs. human DA Spearman r falls below the 0.3threshold for 3of 5low-resource pairs tested: ha = 0.115 (95% CI [0.102, 0.127], n = 23983), ps = 0.295, zu = 0.247 — versus an en-de benchmark of $\approx$ 0.57. To trust this more: extend beyond the five pairs with real WMT DA/MQM human judgments; current coverage is a hard ceiling on what we can check.
- **Source-side triggering (Sec. 5.1) shows no signal from cheap features.** Five feature families (length, OOV proxy, LLM-rated idiomaticity, parse depth, domain markers) predict low quality at AUC 0.513–0.564 against a chance baseline of 0.5. One promising-looking pooled correlation (OOV rate, pooled Pearson $r = -0.33$) evaporates to -0.045–0.021 once controlled for language — a genuine Simpson's paradox, not noise. To trust this more: test a real domain classifier and a frequency-list OOV measure (not a tokenizer-fragmentation proxy) before concluding triggering is a dead end rather than an implementation gap.

The central joint-necessity claim: confounded, not resolved

- **Phase 3** (Aya-101, en-de/en-ha, n = 100/pair): tool-alone beat every other condition on both pairs, and the CoT-vs-tool gap differed significantly by resource level (paired-bootstrap p = 0.004–0.024 across metrics). But Aya-101 is not conversation-tuned, and its CoT chain produced byte-identical draft/refine/proofread output in 80–86out of 100sentences depending on pair/condition. The "CoT hurts" result is real in the sense that it's what happened, but is confounded with architecture, not a clean test of reasoning per se.
- **Phase 4** (Qwen2.5-32B, en-de only, n = 100, 4iterations): neither self-critique (condition A) nor external-QE refinement (condition B) peaked early and declined; both improved gradually to iteration 4/2(gain 0.011/0.01 COMET-KIWI from iteration 1 to peak). This matches what Feng et al. actually report for TEaR, not the early-peak hypothesis we set out to test (that pattern belongs to a different paper). Secondary finding worth a follow-up: condition A froze

only 31/100sentences by iteration 4 vs. condition B's 48/100by iteration 3, and A's trajectory is smoother — span-level feedback (even self-generated) may matter more than feedback source (self vs. external). To trust this more: rerun condition B with XCOMET (span-level) instead of scalar COMET-KIWI to isolate granularity from source.

- **Phases 5–6**: two candidate replacement models for the low-resource arm were ruled out with direct evidence, not assumption. InkubaLM (0.4B params) is a capability floor — incoherent even in English regardless of prompting. BLOOMZ needed 9mechanical fixes to reach a fair test, and then failed on resource-exposure grounds, not reliability grounds (see Phase 6 below). **Net: the joint-necessity question is still untested on a model without a disqualifying confound.**

Tangential but new: a pretraining-exposure floor

- **Phase 6**: BLOOMZ's clean-generation rate (l1_baseline, manually categorized, $n = 10$/pair) tracks its ROOTS pretraining byte count monotonically across four orders of magnitude: en-fr (208242620434 bytes) → 100% clean; en-sw (236482543 bytes) → 50%; en-xh (14304074 bytes) → 0%; en-zu (8511561 bytes) → 0%. Below some exposure level the model doesn't attempt the target language at all — it defaults to echoing the English source (comprehension is intact; generation is not) — rather than degrading gracefully. To trust this more: only 4 points on the curve; need intermediate-exposure languages (e.g. Yoruba, ∼90M bytes) to call it an actual regression rather than an eyeballed trend.
- Cross-checked the same question for Phase 3's Aya-101/mT5: German has 347B tokens in mC4 vs. Hausa's 0.2B ($\sim 1735\times$) — so, unlike BLOOM's zero-exposure German, Phase 3's en-de anchor is legitimate for the model actually used there. Same question, opposite answer for the two models — worth checking per-model going forward, not assuming either way.

Open decision

Three ways to get an unconfounded joint-necessity result:

1. **Keep searching for an open model that is both conversation-tuned and covers the target languages.** Two candidates already ruled out with evidence (9fixes deep, in BLOOMZ's case). No known third candidate identified yet; another search cycle risks another multi-fix debugging odyssey with no guaranteed payoff, but preserves an all-open-source stack.
2. **Split the two roles**: an API model for reasoning/orchestration, a Church-fine-tuned NLLB model queried as the tool, rather than asking one model to both translate and reason. This sidesteps Phase 6's exposure-floor problem entirely (NLLB is purpose-built and trained at scale on exactly these low-resource pairs, unlike a general-purpose LLM's incidental exposure), and arguably tests the theory's actual claim (reasoning *and* tool use, not one model doing both) more faithfully than the current single-model design. Costs real API spend and departs from the fully-open constraint.
3. **Narrow scope**: run the clean joint-necessity test on one pair BLOOMZ can already generate reliably (en-fr, 100% clean per Phase 6), and treat low-resource generalization as explicit future work rather than this paper's claim.

Leaning toward option 2 — it resolves Phase 6's confound directly rather than continuing to search for a single model that clears it. Option 3 is the fallback if preserving an open-only stack matters more than resolving the question now.

What changes in the paper draft

- **Sec. 5.3 (QE-in-the-loop)**: needs an explicit caveat — the proposed mechanism doesn't work for 3/5tested low-resource pairs; reframe as an open problem, not a solved detail.
- **Sec. 5.1 (triggering)**: same treatment — cheap source-side features show no signal; needs either a caveat or a note that only crude implementations have been ruled out.
- **Sec. 1 (motivation)**: add the pretraining-exposure-floor finding (Phase 6) as a distinct failure mode from reasoning/tool insufficiency — "low-resource" can mean "the base model never learned to generate this language at all," independent of any agentic scaffolding.
- **Core joint-necessity claim**, wherever stated: needs a "confounded/untested cleanly" framing — not silence, not overclaiming from Phase 3's suggestive-but-confounded numbers.

Appendix: debugging record (compressed)

- **Aya-101 (Phase 3)**: T5 seq2seq, no chat template; multi-turn context handled by flattening to a role-tagged transcript. Worked for single-turn conditions; degenerated for the 4-step CoT chain (see verbatim-repeat numbers above).
- **Refine-step scoring bug (Phase 4)**: Qwen2.5-32B's refine responses were often full English explanations ("Given the feedback. . . the revised sentence would be:. . . "), and the entire response was being scored as the translation — produced a fake collapse from 0.71 → 0.53 COMET-KIWI. Fixed with an explicit "output only the translation" instruction plus a post-hoc extractor, validated against the original buggy data (287/290 cleaned correctly) before re-running.
- **BLOOMZ/InkubaLM (Phase 5)**: 9fixes in sequence — missing pre-download, an outdated KV-cache code path in InkubaLM's custom model class, partial gated-repo caching, a context-window miscalculation, role-tag framing that made both models echo prompt boilerplate instead of translating, a completion-cue removal that caused empty output, a misdiagnosed repetition-penalty fix, a cue-induced short-answer truncation bug, and finally a genuine reliability problem (inconsistent correct-translation-vs.-source-echo behavior) that was not further fixable. An earlier full run that looked like an interesting contradiction of Phase 3 was explicitly retracted once these bugs were understood — flagged as uncitable in `phase_5_summary.md` rather than left standing.
- **ROOTS-table correction (Phase 6)**: a secondary characterization claimed Xhosa/Zulu were entirely absent from BLOOM's pretraining data. Checked the primary source (BLOOM paper, Table 1) directly — both are present with real, if small, byte counts (14304074 / 8511561 bytes respectively). The claim that German has zero presence, however, held up on the same check — and turned out to be the more consequential finding (Phase 3's/Phase 5's en-de anchor question, resolved oppositely for the two models involved).